

Document information

Info	Content
Keywords	NFC, Android, DTA APK
Abstract	This document provides information on how to use NFC DTA APK on Android platform to validate NFC Forum compliance test cases.

Revision history		
Rev	Date	Description
1.0	2024-09-25	Initial version
1.1	2025-03-14	Updated DTA version to CR14_DTA_1.3
s1.2	2026-01-28	Updated DTA version to CR14_DTA_1.4
1.3	2026-02-26	Updated DTA version to CR14_DTA_1.5 with reader mode annotation fix.
1.4	2026-06-25	Updated DTA version to CR14_DTA_1.6 with fix for the DTA crash 130 KB NDEF write.

Table of Contents

1. Introduction	4
2. Scope.....	4
3. Architecture of NFC DTA APK.....	4
3.1 Testing Scope	5
4. NFC DTA APK setup	5
4.1. Running DTA APK with options	9
4.2.1 NXP_DTA_UI_SCR_SCENERIO_01: Default screen.....	9
4.2.2 NXP_DTA_UI_SCR_SCENERIO_02: Listen mode selection.....	11
4.2.3 NXP_DTA_UI_SCR_SCENERIO_02: Poll(Reader) mode selection.....	12
5. References	13
6. Legal Information.....	13
6.1. Disclaimers	13
6.2. Trademarks	13

1. Introduction

Device Test Application (DTA) that a vendor can integrate in an NFC Forum Device to ensure that the Implementation/Device Under Test (IUT/DUT) can be tested against the NFC Digital Protocol Technical Specification [DIGITAL], NFC Forum Type 2-5 Tag Operation Specifications [TnTOP], NFC Forum Analog RF.

DTA APK is designed to work with NCI based NFC chipsets. This setup guide provides the detailed directions about setting up NFC DTA apk for NFC Forum Compliance Testing of Implementation Under Testing (IUT) or Device Under Testing (DUT).

2. Scope

This document is written considering NFC DTA apk setup guidelines to perform the NFC Forum compliance validation of Implementation Under Testing (IUT) or Device Under Testing (DUT).

3. Architecture of NFC DTA

Figure 1 shows the architecture of NFC DTA.

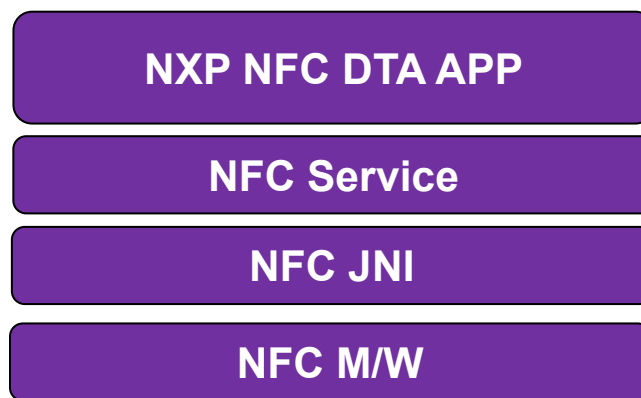


Figure 1: NFC DTA Architecture

NFC DTA supported Features:

A NFC device may support one or more communication technologies: Type A, B and F, in both Poll & Listen modes.

3.1 Testing Scope

- NFC Forum Digital protocol test cases.
- NFC Forum T2T, T3T, T4T & T5T test cases.

4. NFC DTA APK setup

Download binaries & go to the below mentioned path.

build_image\image\NXPDTA\

Following commands need to be executed on the terminal to install the apk & supporting files.

```
adb root
```

```
adb remount
```

```
adb push NXPDTA /system/app/
```

```
adb reboot
```

Note: For Android 17 CR14_DTA_1.5 and later, the DTA package must be pushed to **/system/app**. For earlier DTA versions (CR14_DTA_1.1 to CR14_DTA_1.4), it should be pushed to **/vendor/app**.

After updating the required file, the “NXP DTA” appears in the main menu as shown in Figure 2.



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Figure 2: DTA APK installed in Android.

Configuration changes before running NFC forum test cases:

- pull libnfc-nci.conf from the device using `adb pull /product/etc/libnfc-nci .`
- Modify parameter value `ISO15693_SKIP_GET_SYS_INFO_CMD` from 0 to 1.s
- Push modified conf file in the device using `adb push libnfc-nci.conf /product/etc/`
- Reboot the device using `adb reboot`.
- Due to the ASOP changes introduced in MW 17.07.00, DTA must be enabled after every reboot before launching DTA by executing the following ADB command
`adb root`
`adb remount`
`adb shell cmd nfc configure-dta enable`

Before running DTA APK

Switch ON the NFC service option in Settings, Settings->Connected devices-> Connection preferences -> NFC as ON.

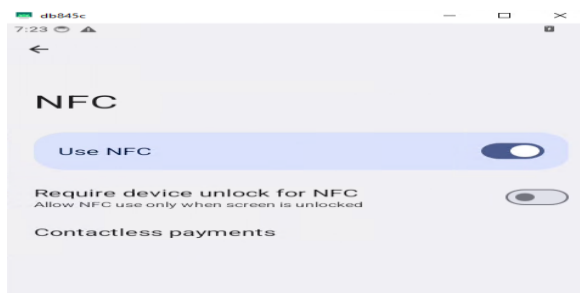


Figure 3: NFC service ON in settings.

Once after NFC is ON in figure 3, select Contactless payments & it will go to Default wallet app screen. In this Select NXP DTA and come back to home screen & launch NXP DTA application.

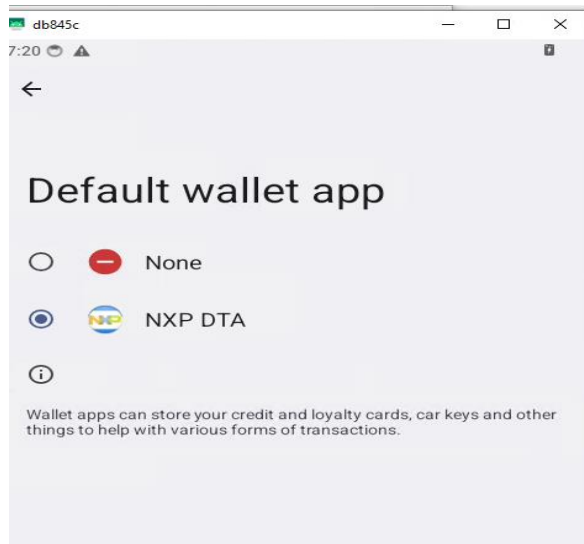


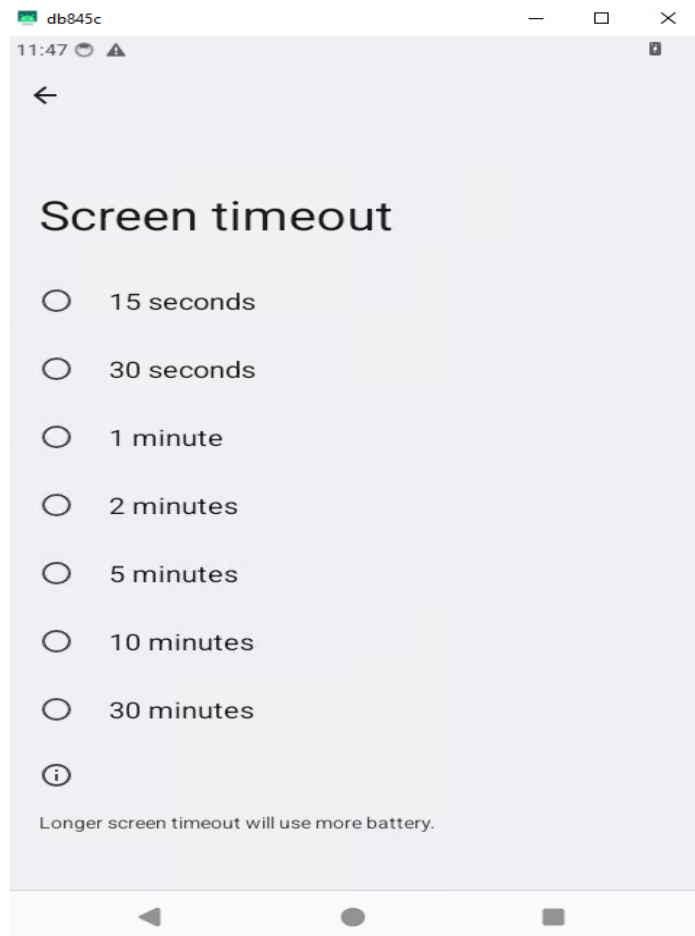
Figure 4: Default wallet app

Screen time out settings.

Screen time out should be updated in the IUT settings to avoid the DTA RF signal loss. Because once the device goes to sleep mode, immediately RF will be stopped from device, to avoid this device screen timeout should be increased to 30 minutes or device should be powered.

The following path can be used for updating the screen timeout setting.

Main menu-> Settings -> Display -> Screen timeout -> select 30 minutes.



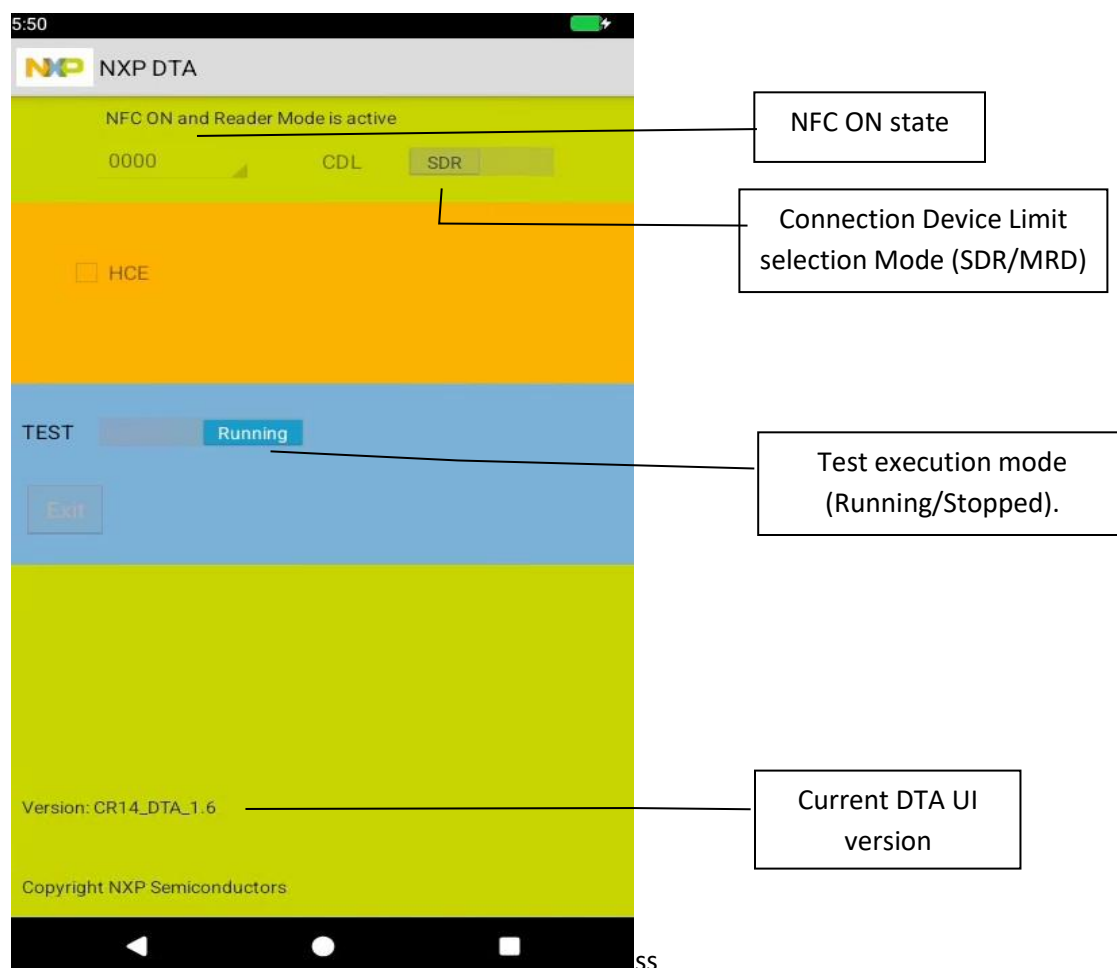
4.1. Running DTA APK with options

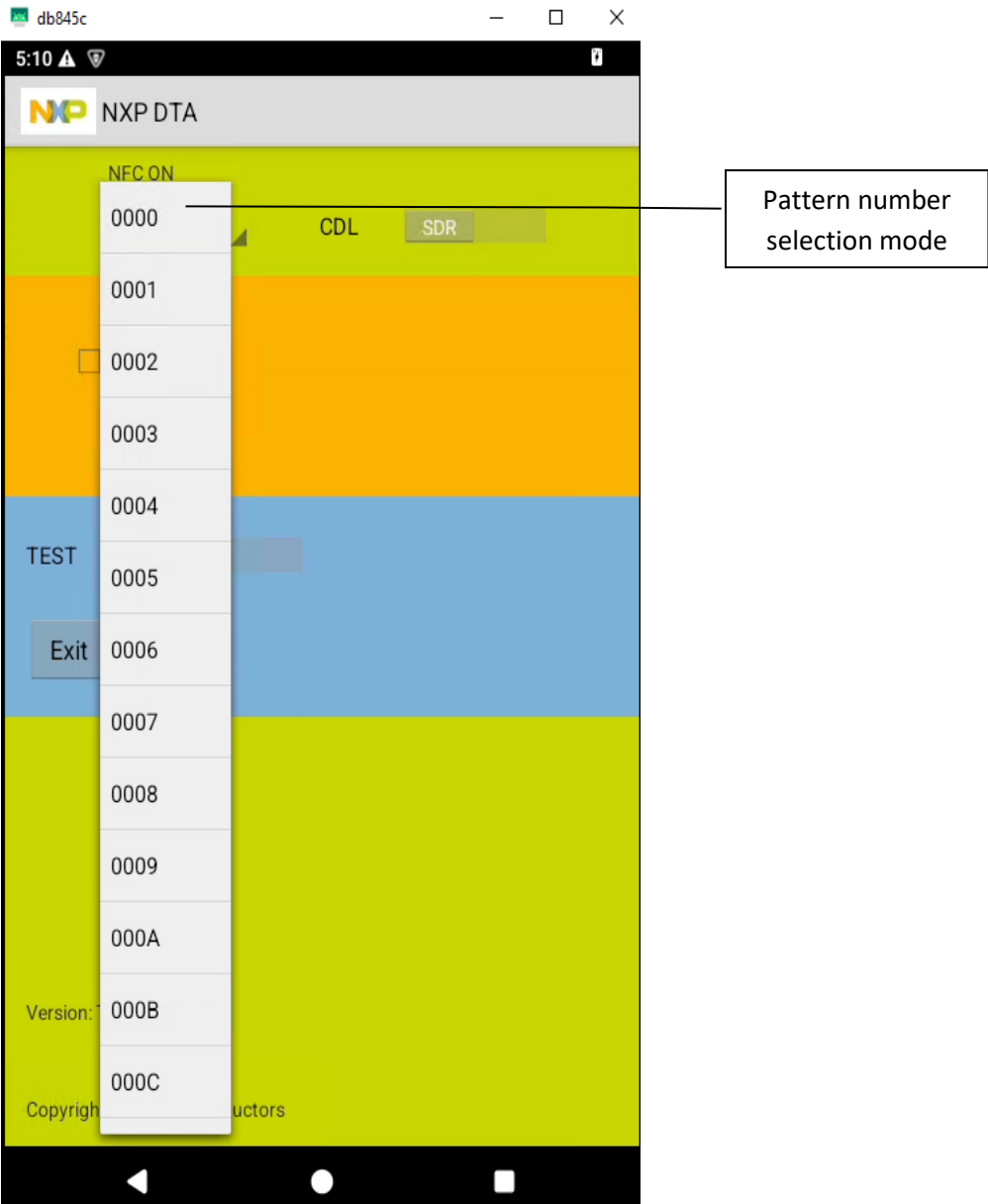
4.2.1 NXP_DTA_UI_SCR_SCENERIO_01: Default screen

The default screen is loaded as soon as the application is launched. By default, NFC ON and the pattern number will be set to “0000” in multi option. Connection Device Limit is set to default Single Device Detection (SDR).

The **current status** of the application is Stopped. To run test cases, select switch button from Stopped to Running. Exit button to exit the test application.

Copyright and DTA UI Version are showed in the bottom.

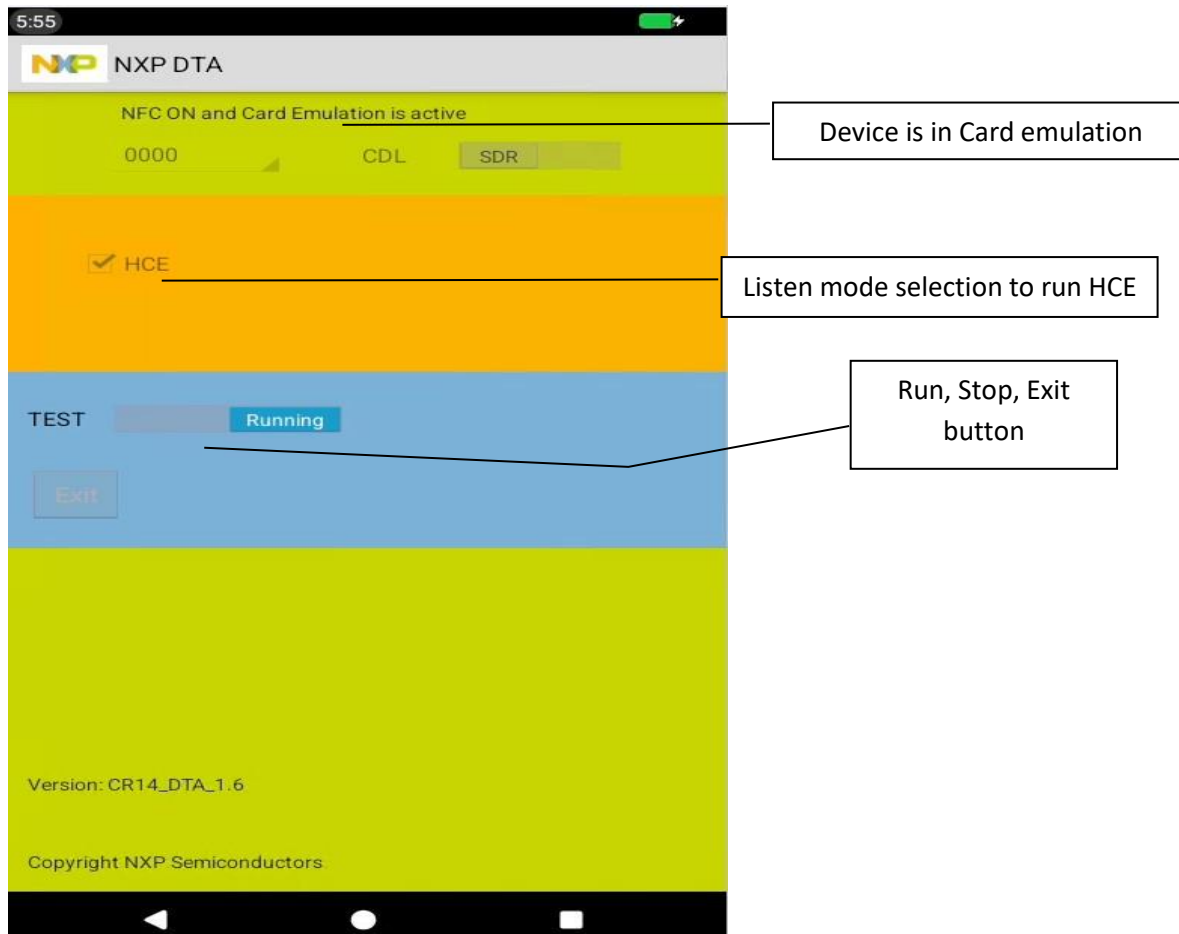




4.2.2 NXP_DTA_UI_SCR_SCENERIO_02: Listen mode selection

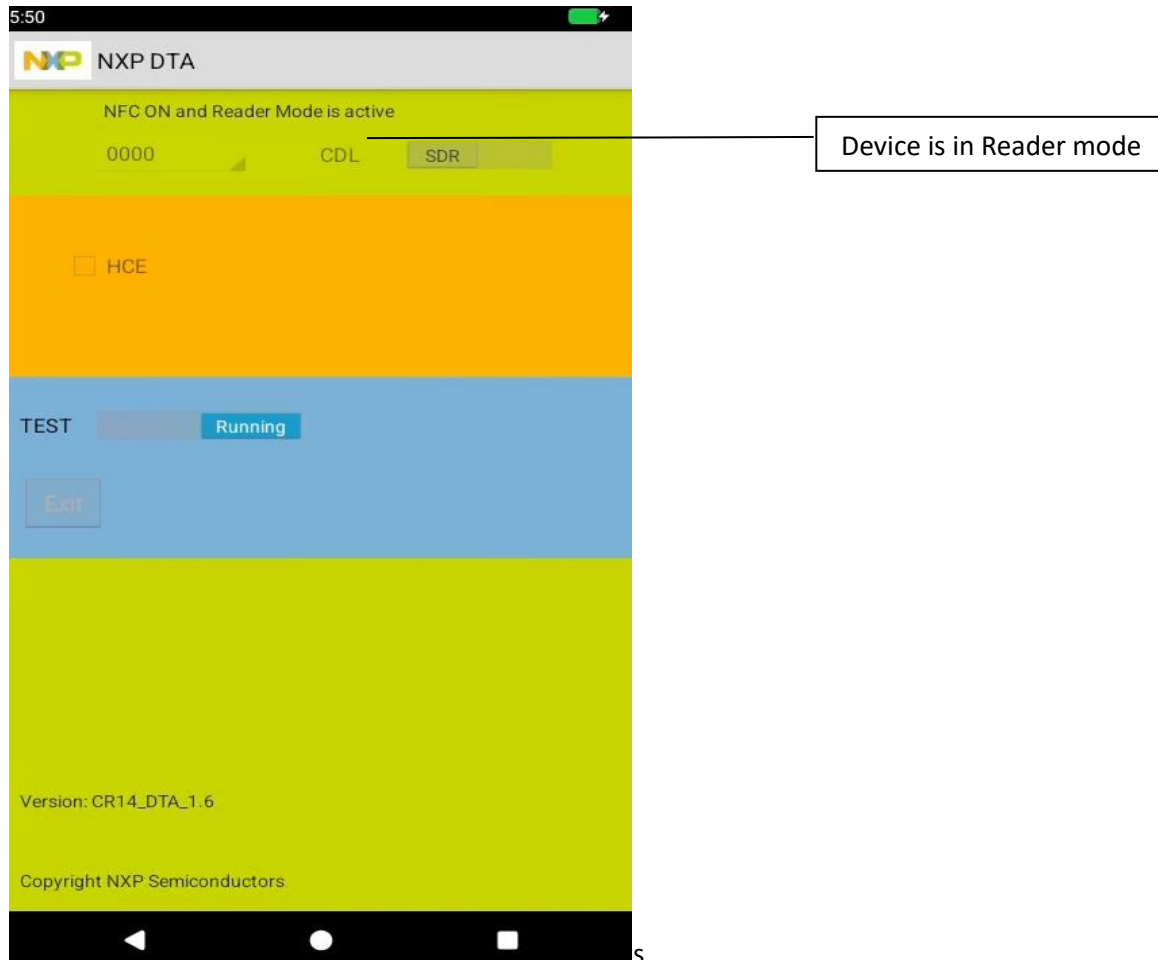
This screen is similar to “NXP_DTA_UI_SCR_SCENERIO_01” screen with the changes shown based on the user selection.

To run Host Card Emulation test cases (Listen mode) need to select HCE check box before test running test case. After this start test case, test execution mode changes from Stopped to **Running** mode & on the top of the application it will display message “NFC ON and Card Emulation is active”.



4.2.3 NXP_DTA_UI_SCR_SCENERIO_02: Poll(Reader) mode selection

First select the pattern number & Connection Device Limit (SDR/MDR) and then run the test case. The current status will be changed to first **Running** and display message “NFC ON and Reader Mode is active” on top of the screen.



5. References

- None

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